

Spatial Logistic Gaussian Processes based surrogates for spatially dependent discrete outputs: modelling, uncertainty quantification, and data acquisition.

Athénaïs Gautier

DTIS, ONERA, Université Paris-Saclay, 91120, Palaiseau, France.

Contributing authors: athenais.gautier@onera.fr;

Abstract

Discrete and categorical responses arise in many modern spatial experiments, but most design methodologies focus on mean behaviour or binary classification. We propose Spatial Logistic Gaussian Processes (SLGPs) as general-purpose surrogates for spatially varying discrete distributions. The SLGP representation yields a full posterior on the probability mass function at any location, allowing principled uncertainty quantification and distribution-aware loss functions. We derive acquisition strategies that select new samples based on the current uncertainty. The approach is evaluated on real and synthetic datasets and compared to multi-class GP classification. Our results show that SLGPs provide substantially more accurate spatial PMF reconstruction in small-data regimes and produce coherent uncertainty estimates, making them attractive tools for experiment design and sequential sampling in settings requiring distribution-level fidelity.

Keywords: Spatial statistics, Discrete-valued fields, Distributional learning, Non-parametric modelling, Uncertainty quantification

1 Introduction

Discrete-valued outputs arise naturally in a wide range of scientific and engineering applications including remote sensing [1], ecological abundance modelling [2, 3], epidemiology [4], medicine [5], and finance [6].

In many such settings, the objective is not merely to predict a single class label or expected count, but rather to model the full conditional distribution of the response as a function of spatial or environmental covariates. Access to the full conditional distribution (rather than a point prediction alone) is essential when decisions must be taken under uncertainty, when data are collected sequentially, or when experiments are costly and require adaptive design strategies [7, 8]. Reliable uncertainty quantification becomes a central modelling requirement, both to assess predictive confidence and to guide principled acquisition criteria.

A broad range of methods has been proposed for discrete-output modelling. Classical approaches include multinomial and ordinal regression models [9, 10], but may be inadequate in the presence of complex non-linear effects or high-dimensional covariates. Tree-based methods such as random forests and gradient boosting machines provide flexible non-linear alternatives [11–13]. While these models have achieved strong empirical performance in multiclass classification and discrete prediction tasks, uncertainty quantification in these models is typically heuristic, which limits their use in sequential design. Neural-network-based classifiers, especially Bayesian neural networks and deep ensemble approaches attempt to address uncertainty quantification [14–16], however, posterior approximations in Bayesian neural networks are often computationally demanding or crude, and ensemble-based uncertainty measures do not generally correspond to a well-defined posterior distribution. Finally, Gaussian process (GP) classifiers constitute an attractive compromise when data efficiency and uncertainty quantification are central concerns, as they provide a Bayesian framework for learning smooth input-dependent class probabilities and naturally support sequential design strategies, with early contributions including information-based criteria and uncertainty sampling and more recent work leveraging predictive entropy and expected information gain [17–23].

Nevertheless, most existing approaches primarily target prediction of a single discrete label. In several applications, however, the primary quantity of interest is the full vector of class probabilities, together with uncertainty on these probabilities themselves. This situation arises naturally in compositional data analysis [24], as well as in settings where probabilities are propagated through downstream models, compared across spatial locations, or used directly within acquisition functions.

Building on the Spatial Logistic Gaussian Process (SLGP) framework originally developed for conditional density estimation [25], this work shows that the same construction naturally extends to finite-support outputs. The resulting model provides a coherent Bayesian framework for learning spatially varying probability mass functions, quantifying uncertainty on these distributions, and guiding adaptive data acquisition.

The main contributions of this work are threefold. First, a discrete-output formulation of SLGPs is introduced, and the original continuous construction is shown to naturally extend to finite-support responses. Second, previous work on SLGPs is extended through the introduction of structured trend components, allowing prior knowledge and support constraints to be incorporated into the model. Third, active-learning strategies based on posterior uncertainty are investigated and illustrated on both synthetic and real-data examples.

The remainder of the paper is organised as follows: Section 2 recalls the SLGP construction and inference, with emphasis on discrete and ordinal outputs. Section 3 discusses uncertainty quantification and acquisition strategies enabled by SLGP-based surrogates. Section 4 presents analytical and real-data applications.

2 Spatial Logistic Gaussian Processes: From the continuous to the discrete setting

This section introduces the Spatial Logistic Gaussian Process framework and explains how the original continuous-output construction can be naturally extended to discrete and ordinal responses. We start by recalling the standard SLGP formulation for spatially dependent density estimation, and then show how it naturally adapts to finite-support outputs, yielding a coherent probabilistic framework for modelling spatially varying probability mass functions, and discuss links to multinomial logistic regression and multi-class GP.

2.1 SLGP in the continuous output setting

The Spatial Logistic Gaussian Process (SLGP), developed in [25] as a spatial extension of Logistic Gaussian Process models [26, 27] and related to the framework of [28], defines a non-parametric prior useful for modelling families of probability densities indexed by covariates.

More precisely, let $T_{\mathbf{x}} \in \mathcal{T}$ denote a continuous random variable whose distribution depends on $\mathbf{x} \in D$ (e.g., spatial location or covariates) and needs to be modelled and estimated. The response space $\mathcal{T} \subset \mathbb{R}^{d_T}$ is assumed compact and convex with $d_T \geq 1$, and the indexing domain $D \subset \mathbb{R}^{d_X}$ satisfies $d_X \geq 1$. The SLGP construction leverages a latent *well-behaved* Gaussian Process (GP) $(Z_{\mathbf{x},t})_{(\mathbf{x},t) \in D \times \mathcal{T}}$ and applies the *spatial logistic density* transformation to it:

$$\Psi[Z](\mathbf{x}, t) = Y_{\mathbf{x},t} = \frac{e^{Z_{\mathbf{x},t}}}{\int_{\mathcal{T}} e^{Z_{\mathbf{x},u}} du} \text{ for all } (\mathbf{x}, t) \in D \times \mathcal{T} \quad (1)$$

At any fixed $\mathbf{x}$, $\Psi[Z](\mathbf{x}, \cdot)$ hence returns a random function that is, by construction, positive and integrates to one. Hence, a SLGP can be (informally) seen as a field of random pdfs. The precise mathematical nature of the objects involved (whether in terms of random measures or densities, and fields thereof) requires careful treatment, for which we refer the reader to [25].

Structured latent representation and trend components

To enhance modelling flexibility and incorporate prior knowledge, we extend the original formulation by introducing an explicit structured component in the latent field:

$$Z_{\mathbf{x},t} = \mu(\mathbf{x}, t) + Z_{\mathbf{x},t}^{(0)}, \quad (2)$$

where $\mu(\mathbf{x}, t)$ is a deterministic trend function and $Z^{(0)}$ is a zero-mean Gaussian process capturing residual variability.

This decomposition separates large-scale or physically motivated effects from stochastic deviations. The structured component μ can encode expert knowledge, known asymptotic behaviour, or interpretable parametric structure, while Z_0 retains the non-parametric flexibility of the SLGP.

In the present implementation, μ is specified a priori and not jointly estimated with Z_0 . This choice avoids over-parametrization, enhances numerical stability, and preserves the coherence of posterior uncertainty quantification inherited from the Gaussian process prior.

Implementation of the continuous SLGP

Posterior inference under an SLGP prior raises two main challenges: the infinite-dimensional nature of the latent Gaussian field and the repeated evaluation of the normalisation functional. In particular, the normalising constant $\int_{\mathcal{T}} e^{Z_{\mathbf{x},u}} du$ must be computed for many input locations $\mathbf{x}$ during posterior optimisation or sampling, which can quickly become computationally prohibitive.

To obtain a tractable implementation, we combined two complementary strategies.

Finite-rank representation of the latent field. We adopt a finite-rank representation of the latent process, for instance via inducing-point approximations or Fourier-type bases such as Random Fourier Features, and write

$$Z(\mathbf{x}, t) := \sum_{i=1}^p \varepsilon_i f_i(\mathbf{x}, t) = \boldsymbol{\varepsilon}^\top \mathbf{F}(\mathbf{x}, t), \quad (3)$$

where $\boldsymbol{\varepsilon}$ follows a multivariate normal prior and $\mathbf{F} = (f_i)_i$ denotes a collection of known basis functions. Under this representation, inference reduces to a finite-dimensional problem in $\boldsymbol{\varepsilon}$ and basis evaluations can be precomputed and reused across repeated $\mathbf{x}$ values, significantly reducing computational overhead.

Quadrature approximation of the normalisation. We approximate the normalising integral by numerical quadrature. Let $(u_j)_{j=1}^m \subset \mathcal{T}$ and $(w_j)_{j=1}^m \subset \mathbb{R}_+^m$ respectively denote the quadrature nodes and weights. Then we use the following approximation:

$$\int_{\mathcal{T}} e^{Z_{\mathbf{x},u}} du \approx \sum_{j=1}^m w_j e^{Z_{\mathbf{x},u_j}}. \quad (4)$$

Posterior uncertainty quantification and posterior inference is then performed by generating samples $\boldsymbol{\varepsilon}^{(1)}, \dots, \boldsymbol{\varepsilon}^{(N)}$ from an approximation of the posterior distribution $p(\boldsymbol{\varepsilon}|\mathcal{D}_n)$. Depending on the inference scheme, these samples are either obtained directly using MCMC (via the NUTS algorithm implemented in Stan) or drawn from a Laplace approximation around the MAP estimator. These posterior draws are then propagated through the logistic transformation to obtain samples from the posterior distribution of the density field. Detailed implementation aspects are discussed in [29], and made available in the accompanying software [30].

Remark 1 Here and in the rest of the paper, the notation ε and variations thereof is destined to the random variables, while ϵ is used for samples and realisations.

2.2 A principled extension to discrete (and ordinal) responses.

The quadrature mechanism naturally provides an extension of the SLGP framework to discrete-output modelling. Indeed, when the response takes values on a finite support $\{1, \dots, K\}$, choosing quadrature nodes equal to this support renders the normalisation exact. The SLGP can then be identified to a random probability mass function (pmf):

$$Y_{\mathbf{x},k} := \frac{e^{Z_{\mathbf{x},k}}}{\sum_{j=1}^K e^{Z_{\mathbf{x},j}}} \quad \text{for all } \mathbf{x} \in D \text{ and } k = 1, \dots, K \quad (5)$$

This discrete-output SLGP is not a separate construction, but a particular instance of the very same spatial logistic Gaussian framework in which the reference measure becomes atomic rather than Lebesgue.

Remark 2 The discrete-output formulation can be viewed as a simplified special case of the original SLGP framework. In the continuous setting, the logistic transformation involves a normalization integral over the response domain, which requires numerical quadrature $\int_{\mathcal{T}} e^{Z_{\mathbf{x},u}} du \approx \sum_{j=1}^m w_j e^{Z_{\mathbf{x},u_j}}$. This makes the inference approximate, and contributes significantly to the computational cost of inference. In contrast, here this integral reduces to a finite sum, $\sum_{j=1}^K e^{Z_{\mathbf{x},j}}$. As a result, no approximation of the numerical integration is required. The resulting model retains the probabilistic interpretation and uncertainty quantification capabilities of SLGPs while significantly simplifying posterior computations.

Combining the trend (2) and the finite-rank representation (3), the discrete-output SLGP likelihood of class k ($1 \leq k \leq K$) at location $\mathbf{x} \in D$ takes the explicit form:

$$P(T_{\mathbf{x}} = k|Y) = P(T_{\mathbf{x}} = k|\boldsymbol{\varepsilon}) = Y_{\mathbf{x},k} = \frac{\exp \{ \mu(\mathbf{x}, k) + \boldsymbol{\varepsilon}^\top \mathbf{F}(\mathbf{x}, k) \}}{\sum_{j=1}^K \exp \{ \mu(\mathbf{x}, j) + \boldsymbol{\varepsilon}^\top \mathbf{F}(\mathbf{x}, j) \}}. \quad (6)$$

Remark 3 Although the discrete-output setting only requires probabilities on the finite support $\{1, \dots, K\}$, the dependence on the input variable x must remain continuous in order to perform prediction at arbitrary locations. This requirement is particularly important in the sequential design setting considered later, where acquisition functions must be optimized over the entire input space. For this reason, the finite-rank representation is retained from the original SLGP framework, yielding a continuous surrogate model that can be queried at previously unobserved inputs.

Data integration and class prediction with the discrete-output SLGP

Inference can now be performed within the discrete-output SLGP framework. Let $\mathcal{D}_n = \{(\mathbf{x}_i, t_i)\}_{i=1}^n \in (D \times \{1, \dots, K\})^n$ denote observed data, where each $t_i \in \{1, \dots, K\}$ is a realisation of the pairwise independent random variables $T_{\mathbf{x}_i}$.

The relationships between the different quantities involved in the model are summarized in Figure 1. In particular, inference is performed on the latent coefficient vector $\boldsymbol{\varepsilon}$, from which posterior samples of the probability field $Y_{\mathbf{x},k}$ and posterior predictive probabilities are obtained.

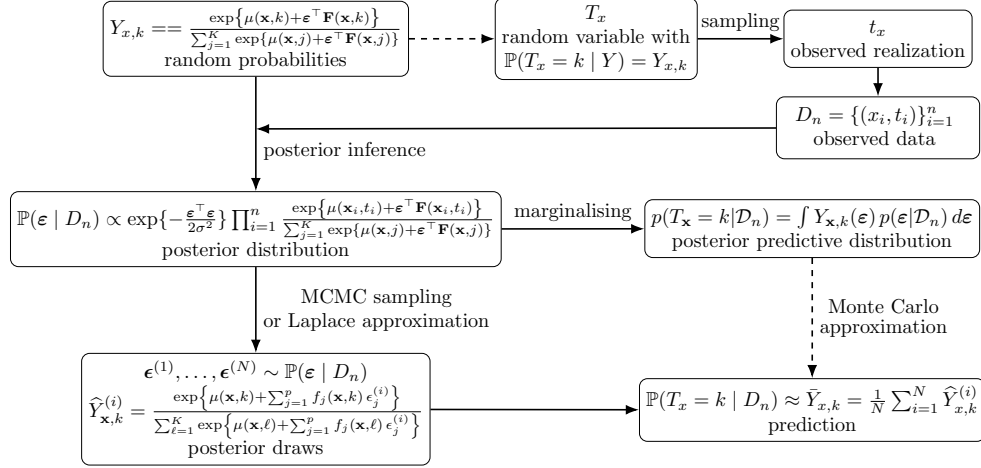


Fig. 1: Overview of the mathematical objects involved in the SLGP model and their relationships, from the SLGP (based on a latent finite rank Gaussian Process) to posterior predictive probabilities.

Posterior inference targets the distribution of the latent coefficients ϵ conditionally on the observations, yielding the posterior:

$$P(\epsilon | T_1 = t_1, \dots, T_n = t_n) \propto \phi_p(\epsilon) \prod_{i=1}^n \frac{\exp\{\mu(\mathbf{x}_i, t_i) + \epsilon^\top \mathbf{F}(\mathbf{x}_i, t_i)\}}{\sum_{j=1}^K \exp\{\mu(\mathbf{x}_i, j) + \epsilon^\top \mathbf{F}(\mathbf{x}_i, j)\}}, \quad (7)$$

where ϕ_p the pdf of the p -variate standard normal distribution. As in the continuous case, posterior uncertainty quantification is then performed by generating samples $\epsilon^{(1)}, \dots, \epsilon^{(N)}$ from the posterior distribution, whether by directly using MCMC or by leveraging a Laplace approximation. Detailed implementation aspects for this sampling are discussed in Gautier [29] and made available through the accompanying software package [30].

Each posterior sample induces a realization of the probability field through

$$\hat{Y}_{\mathbf{x},k}^{(i)} = \frac{\exp\left\{\mu(\mathbf{x},k) + \sum_{j=1}^p f_j(\mathbf{x},k) \epsilon_j^{(i)}\right\}}{\sum_{\ell=1}^K \exp\left\{\mu(\mathbf{x},\ell) + \sum_{j=1}^p f_j(\mathbf{x},\ell) \epsilon_j^{(i)}\right\}}, \quad i = 1, \dots, N. \quad (8)$$

The posterior predictive probability of class k at any input location $\mathbf{x}$ is then obtained by marginalising over the latent parameters:

$$p(T_{\mathbf{x}} = k | \mathcal{D}_n) = \int Y_{\mathbf{x},k}(\epsilon) p(\epsilon | \mathcal{D}_n) d\epsilon, \quad (9)$$

which is approximated using the posterior samples as:

$$p(T_{\mathbf{x}} = k | \mathcal{D}_n) \approx \bar{Y}_{\mathbf{x},k} = \frac{1}{N} \sum_{i=1}^N \hat{Y}_{\mathbf{x},k}^{(i)}. \quad (10)$$

The expression of the discrete-output SLGP in Equation 5 is closely related to widely used models for multiclass prediction. In the following paragraph, we clarify the structural differences between this construction and classical multiclass regression frameworks. For clarity, we align the notation with that of the discrete-output SLGP in order to emphasize both similarities and structural distinctions.

2.2.1 Connection with Bayesian multinomial logistic regression.

In classical multinomial logistic regression, the probability of observing class k ($1 \leq k \leq K$) conditionally on covariates $\mathbf{x} \in D$ is modelled as

$$\pi_k^{multinom.}(\mathbf{x}) = \frac{\exp\{\tilde{\boldsymbol{\epsilon}}_k^\top \tilde{\mathbf{F}}(\mathbf{x})\}}{\sum_{j=1}^K \exp\{\tilde{\boldsymbol{\epsilon}}_j^\top \tilde{\mathbf{F}}(\mathbf{x})\}}. \quad (11)$$

where $\tilde{\mathbf{F}}(\mathbf{x}) \in \mathbb{R}^P$ denotes a vector of covariate-dependent features and $\tilde{\boldsymbol{\epsilon}}_k \in \mathbb{R}^P$ is the coefficient vector associated with class t_k . In a Bayesian formulation, independent Gaussian priors may be assigned to the class-specific coefficients $\tilde{\boldsymbol{\epsilon}}_k$.

Both multinomial logistic regression and the finite-rank discrete-output SLGP rely on a softmax transformation of latent linear predictors, Gaussian priors, and may incorporate deterministic trend components. Beyond these similarities, however, their structural organization differs fundamentally.

In multinomial logistic regression, the feature map $\tilde{\mathbf{F}}(\mathbf{x})$ is shared across classes, while the coefficient vectors $\tilde{\boldsymbol{\epsilon}}_k$ vary with k . In contrast, the discrete-output SLGP employs a single shared random vector $\boldsymbol{\epsilon}$, and class-specific variation is encoded through outcome-dependent basis functions $\mathbf{F}(\mathbf{x}, t_k)$.

Consequently, the multinomial model estimates up to $K \times P$ free parameters, and cross-class dependence arises only through prior specification on these coefficients. By contrast, in the SLGP construction we are working with a p -dimensional subspace instead, and induce structured and input-dependent correlations across classes through the joint basis representation.

Approximate inference schemes developed in related contexts are also relevant. Nested expectation propagation has been proposed for multinomial probit GP classification [31], and data-augmentation strategies have been studied in Bayesian multinomial logistic regression to facilitate posterior sampling in high-cardinality regimes [32]. Given the similarity of the likelihood structures, such techniques may offer promising directions for improving computational efficiency in SLGP-based implementations.

2.2.2 Connection with multiclass Gaussian process classifiers.

In multiclass GP classification, one typically introduces K latent Gaussian processes $\tilde{\mathbf{Z}} := \left(\tilde{\mathbf{Z}}_{\mathbf{x}}^{(k)} \right)_{\mathbf{x} \in D}$, one per class, and defines the likelihood through a softmax transformation:

$$\pi_k^{\text{multiGP}}(\mathbf{x}) = \frac{\exp \left\{ \tilde{\mathbf{Z}}_{\mathbf{x}}^{(k)} \right\}}{\sum_{j=1}^K \exp \left\{ \tilde{\mathbf{Z}}_{\mathbf{x}}^{(j)} \right\}} \quad \text{for any } \mathbf{x} \in D, \text{ and any } k \in \{1, \dots, K\}. \quad (12)$$

The latent processes may be modelled independently or endowed with cross-covariance structures to encode prior correlations across classes.

Given observations $\mathcal{D}_n$, the posterior predictive probability of class k at $\mathbf{x}$ is obtained by marginalising over the latent Gaussian field:

$$p(T_{\mathbf{x}} = k | \mathcal{D}_n) = \int \pi_k^{\text{multiGP}}(\mathbf{x}) p(\tilde{\mathbf{Z}} | \mathcal{D}_n) d\tilde{\mathbf{Z}}, \quad (13)$$

where $\tilde{\mathbf{Z}} = (\tilde{\mathbf{Z}}^{(1)}, \dots, \tilde{\mathbf{Z}}^{(K)})$ collects the latent GPs.

Because the softmax likelihood is non-Gaussian, this integral is analytically intractable and must be approximated. This difficulty is shared with the discrete-output SLGP and reflects the non-conjugacy between the Gaussian prior and the softmax likelihood. However, multiclass Gaussian process classification has been studied extensively over the past decades, resulting in a rich literature on scalable and accurate approximation schemes. Common approaches include Laplace approximation, expectation propagation, and variational inference. A substantial literature addresses scalable inference for multiclass GP classifiers, often through auxiliary-variable constructions designed to recover conditional conjugacy. For instance, Wenzel et al. [33] leverage Pólya–Gamma augmentation in the binary GP setting, while Galy-Fajou et al. [34] introduce a modified softmax likelihood together with an augmentation scheme yielding a conditionally conjugate multiclass GP model. Earlier multinomial probit formulations exploit truncated Gaussian representations to enable Gibbs sampling and variational updates [35, 36].

Structurally, multiclass GP classification differs from the discrete-output SLGP in the way latent dependence across classes is encoded. Standard multiclass GP models associate one latent process $\tilde{\mathbf{Z}}^{(k)}$ with each class and do not intrinsically account for any ordering of the labels. In contrast, the SLGP models a single Gaussian process $Z(\mathbf{x}, t_k)$ on the product space $D \times \{1, \dots, K\}$, allowing neighbouring categories to be related through this shared latent representation. As a result, ordinal information is incorporated naturally into the model, which is particularly relevant when the response corresponds to counts or ordered categories. This distinction also affects the computational scalability: multiclass GP classification scales with the number of latent processes, whereas the discrete-output SLGP maintains a fixed latent dimensionality under the finite-rank construction.

3 Active data acquisition using SLGP

Bayesian modelling is particularly well suited to data acquisition and experimental design, since acquisition criteria can be derived directly from posterior uncertainty [7, 17]. In particular, when the quantity of interest is a predictive distribution rather than a single point estimate, coherent posterior uncertainty quantification becomes essential for guiding informative sampling decisions.

The SLGP framework yields posterior distributions over conditional densities or probability mass functions indexed by $\mathbf{x} \in D$. This distributional structure naturally supports active learning strategies that target uncertainty on the entire predictive pmf field. In the context of GP-based classification, several acquisition criteria have been proposed, typically derived from predictive entropy, mutual information, or posterior variance considerations [20–22]. Below, we briefly review the most commonly used ones and provide the formula used to approximate them within the discrete-output SLGP setting.

- **Predictive entropy (uncertainty sampling)** [20, 21].

The predictive entropy at $\mathbf{x}$ is defined as the Shannon entropy of the posterior mean predictive pmf:

$$a_{\text{Ent}}(\mathbf{x}) = - \sum_{k=1}^K \mathbb{E}[Y_{\mathbf{x},k} | \mathcal{D}_n] \log \mathbb{E}[Y_{\mathbf{x},k} | \mathcal{D}_n]. \quad (14)$$

This criterion favours locations where the predictive distribution is most diffuse, i.e., where classification uncertainty is maximal. It is computationally inexpensive and widely used, but it does not distinguish between model uncertainty and aleatoric uncertainty.

Within the SLGP framework, the posterior expectation $\mathbb{E}[Y_{\mathbf{x},k} | \mathcal{D}_n]$ is approximated using (approximate) draws of the random pmf. Denoting by $\bar{Y}_{\mathbf{x},k}$ the mean of the posterior draws as in (10), we estimate

$$\hat{a}_{\text{Ent}}(\mathbf{x}) = - \sum_{k=1}^K \bar{Y}_{\mathbf{x},k} \log \bar{Y}_{\mathbf{x},k} \quad (15)$$

- **Bayesian Active Learning by Disagreement (BALD)** [22].

BALD selects the location that maximises the mutual information between the next observation and the latent model structure. In the SLGP framework, this mutual information can be written as

$$a_{\text{BALD}}(\mathbf{x}) = H(T_{\mathbf{x}} | \mathcal{D}_n) - \mathbb{E}[H(T_{\mathbf{x}} | Y_{\mathbf{x}}) | \mathcal{D}_n], \quad (16)$$

where $H(T_{\mathbf{x}} | \mathcal{D}_n)$ denotes the Shannon entropy of the posterior predictive distribution.

Using posterior draws $\hat{Y}_{\mathbf{x},k}^{(i)}$ as in (8), a Monte Carlo estimator is

$$\hat{a}_{\text{BALD}}(\mathbf{x}) = - \sum_{k=1}^K \bar{Y}_{\mathbf{x},k} \log \bar{Y}_{\mathbf{x},k} + \frac{1}{N} \sum_{i=1}^N \sum_{k=1}^K \hat{Y}_{\mathbf{x},k}^{(i)} \log \hat{Y}_{\mathbf{x},k}^{(i)}. \quad (17)$$

Unlike predictive entropy, which measures total uncertainty, BALD isolates model uncertainty arising from imperfect knowledge of the model parameters. It therefore favours locations where posterior pmf draws exhibit strong disagreement, i.e., where a new observation is expected to produce the greatest reduction in posterior uncertainty [22].

- **Posterior variance of class probabilities (distributional variance).**

Because SLGP provides a posterior distribution over the entire predictive pmf, a natural measure of epistemic uncertainty is the dispersion of class probabilities under the posterior. We define the total posterior variance at location $\mathbf{x}$ as

$$a_{\text{VarProb}}(\mathbf{x}) = \sum_{k=1}^K \text{Var}(Y_{\mathbf{x},k} | \mathcal{D}_n), \quad (18)$$

where the variance is taken with respect to the posterior distribution over the latent process. This can be interpreted as an ℓ_2 -type uncertainty measure.

Using the notation of (8), a Monte Carlo estimator is given by

$$\hat{a}_{\text{VarProb}}(\mathbf{x}) = \sum_{k=1}^K \frac{1}{N-1} \sum_{i=1}^N \left(\hat{Y}_{\mathbf{x},k}^{(i)} - \bar{Y}_{\mathbf{x},k} \right)^2. \quad (19)$$

This criterion measures dispersion directly in probability space and therefore aligns naturally with the SLGP modelling philosophy, which treats the full conditional pmf as the primary object of inference.

In the following section, we illustrate the SLGP modelling and inference procedures together with the acquisition strategies introduced above. We consider synthetic examples where the ground-truth conditional distribution is known, allowing quantitative assessment of both posterior approximation quality and sequential design performance.

4 Applications

We now illustrate the discrete SLGP framework on both synthetic and real-data applications. The first examples are controlled synthetic benchmarks based on binomial distributions, allowing quantitative assessment of posterior approximation quality and sequential acquisition strategies under known ground truth. We then demonstrate the applicability of discrete SLGPs on a real ecological dataset. In all the applications below, $N = 1000$ posterior draws are used, whether for SLGP or GP-based inference. Preliminary experiments indicated that predictive quantities were stable beyond a few hundred draws.

The full experimental code, including data generation, model fitting, and acquisition procedures, is available in the accompanying GitHub repository [37].

4.1 Case 1: Binomial distribution with fixed n and varying p

We first consider a controlled one-dimensional setting in which the conditional distribution is analytically known, allowing quantitative evaluation of approximation accuracy and design efficiency.

4.1.1 Data-generating process and SLGP modelling

We fix $n_{Bin} = 10$ and let the input be one-dimensional, $\mathbf{x} = p \in (0, 1)$. We therefore denote the data-generating random variables by $T_{\mathbf{x}} = T_p$ in this section. Observations are generated according to $T_p \sim \text{Bin}(10, p)$, so that the support of T_p is $\{0, \dots, 10\}$ and the number of categories is $K = 11$.

The conditional probability mass function is

$$\mathbb{P}(T_p = k) = \binom{10}{k} p^k (1-p)^{10-k}, \quad k = 0, \dots, 10. \quad (20)$$

This example defines a smooth parametric family of unimodal discrete distributions indexed by p , with varying skewness across the domain. For small p , the distribution is right-skewed, for large p , it is left-skewed, and near $p = 0.5$, it is symmetric. Figure 2 visualizes the resulting probability mass functions as a function of the parameter p and outcome k .

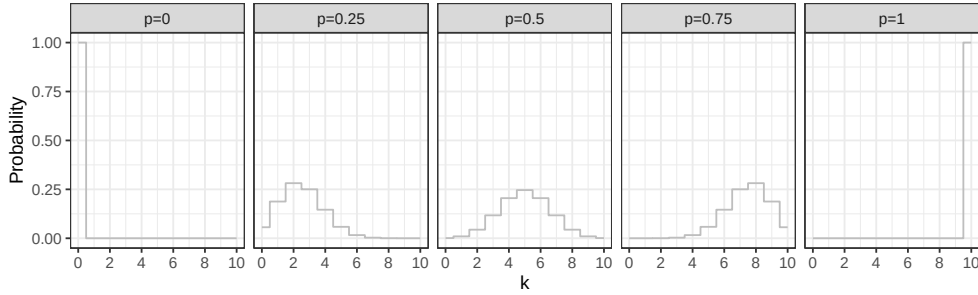


Fig. 2: Ground-truth binomial probability mass function for $n = 10$, displayed for some values of p .

To model this family, we employ a finite-rank SLGP. The basis functions are constructed using a Random Fourier Feature (RFF) approximation of a Matérn-5/2 kernel. We sample 100 frequencies, yielding 200 sine and cosine basis functions. Inference is performed following the implementation described in [29], using both Laplace approximation and MCMC for posterior comparison.

4.1.2 Posterior approximation: MCMC versus Laplace

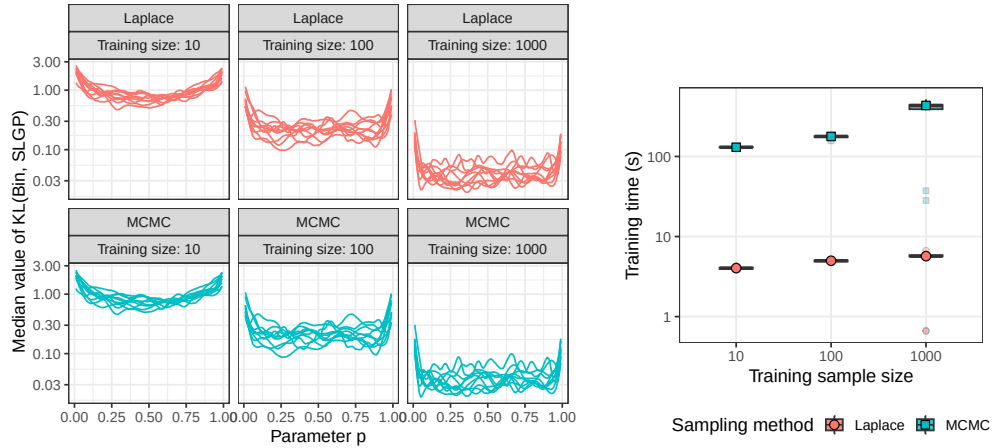
We first compare posterior inference obtained via MCMC sampling with that produced by a Laplace approximation of the posterior over the latent coefficients. Since

the SLGP likelihood is log-concave and the prior Gaussian, the posterior is log-concave, suggesting that a Laplace approximation should be suitable in this setting and is expected to provide a good approximation around the posterior mode. Both approaches are available in the R SLGP package [30], whose implementation relies on the RStan package [38]. For MCMC sampling, the No-U-Turn Sampler (NUTS) algorithm [39] is used.

We therefore begin by comparing the predictive distributions obtained under both inference schemes. To quantify the discrepancy between the SLGP predictive distribution and the ground-truth binomial pmf, we evaluate the pointwise Kullback-Leibler (KL) divergence over a dense grid of $p \in (0, 1)$ for both inference schemes. Recall that for two discrete probability mass functions f and g with common support $\{0, \dots, K\}$, the KL divergence is defined as

$$\text{KL}(f, g) = \sum_{k=0}^K f(k) \log \frac{f(k)}{g(k)}. \quad (21)$$

We perform posterior inference on data sets with $n \in \{10, 100, 1000\}$ observations. The design points p_i are sampled independently from the uniform distribution on $(0, 1)$, and corresponding responses are generated according to $T_p \sim \text{Bin}(10, p)$. Figure 3b reports the KL divergence across a grid of p values under both inference schemes over 10 repetitions, as well as corresponding training time on a laptop. The discrepancy between Laplace and MCMC-based estimates is negligible relative to the approximation error induced by finite data.



(a) Predictive accuracy measured through the Kullback–Leibler divergence.

(b) Computational time required for posterior inference.

Fig. 3: Comparison of Laplace approximation and MCMC inference for the SLGP model. The two approaches yield comparable predictive performance, while the Laplace approximation substantially reduces computational cost.

Given the substantially reduced computational cost of Laplace inference, we adopt it in the subsequent sequential design experiments.

4.1.3 Sequential design of experiments

We now evaluate the acquisition strategies introduced in Section 3 within a sequential design setting. We initialize the model with $n_0 = 10$ observations. Initial design points $p_1, \dots, p_{n_0}$ are sampled independently from the uniform distribution on $(0, 1)$, and corresponding responses are generated according to $T_p \sim \text{Bin}(10, p)$.

At each subsequent iteration, a new design point p_{new} is selected using one of the following strategies: random sampling, predictive entropy, BALD, posterior variance of class probabilities. For each selected p_{new} , we generate 5 independent binomial observations from $\text{Bin}(10, p_{\text{new}})$ and update the SLGP model accordingly. The choice of five observations is primarily motivated by simplicity and is kept fixed throughout the experiments in order to focus on the behaviour of the acquisition criteria. This procedure is carried out for 38 iterations, resulting in a final dataset of 200 observations. We repeat the experiment 10 times for each configuration.

Predictive performance is evaluated through the Integrated Kullback-Leibler divergence. It is defined for two discrete probability mass functions f_p and g_p with common support $\{0, \dots, K\}$ as:

$$\text{IKL}(f, g) = \int_p \text{KL}(f_p, g_p) dp = \int_p \sum_{k=0}^K f_p(k) \log \frac{f_p(k)}{g_p(k)} dp \quad (22)$$

In practice, the integral over p is approximated numerically over a dense grid.

Figure 4 reports the evolution of the median IKL as a function of dataset size across acquisition strategies and 10 repetitions of the experiment.

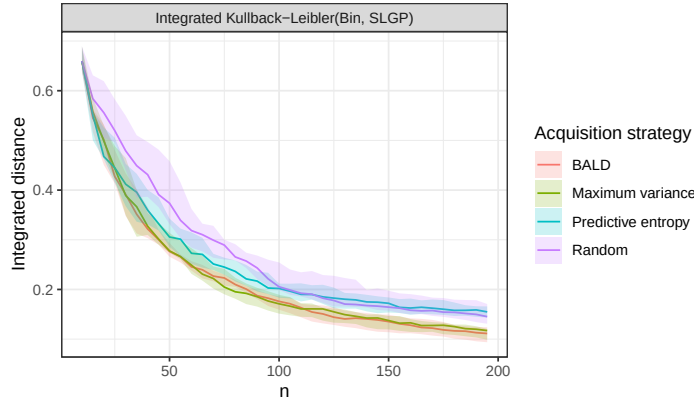


Fig. 4: Median integrated Kullback-Leibler divergence between the SLGP prediction and the ground-truth binomial model as a function of dataset size, under different acquisition strategies. Shaded ribbons indicate the interquartile range (25%-75% quantiles) across independent repetitions.

Random sampling and predictive entropy exhibit comparable behaviour during the early stages of the sequential procedure. However, as the dataset grows, predictive entropy becomes slightly less effective than random sampling. This behaviour is consistent with the fact that entropy-based criteria quantify total predictive uncertainty and therefore cannot distinguish between epistemic uncertainty (due to limited data) and aleatoric uncertainty (intrinsic randomness of the binomial model). Consequently, predictive entropy may continue to select points in regions that are inherently noisy rather than insufficiently explored.

In contrast, BALD and posterior-variance strategies consistently achieve lower approximation error throughout the sequential procedure. Both explicitly target epistemic uncertainty, leading to more informative design points. The posterior-variance criterion slightly outperforms BALD in this setting, although their overall behaviour is close. These results confirm that explicitly isolating model uncertainty substantially improves distributional learning efficiency.

The panel plots in Figure 5 highlight qualitative differences between acquisition strategies by visualising the mean prediction of SLGPs under different training data vs. the actual ground-truth.

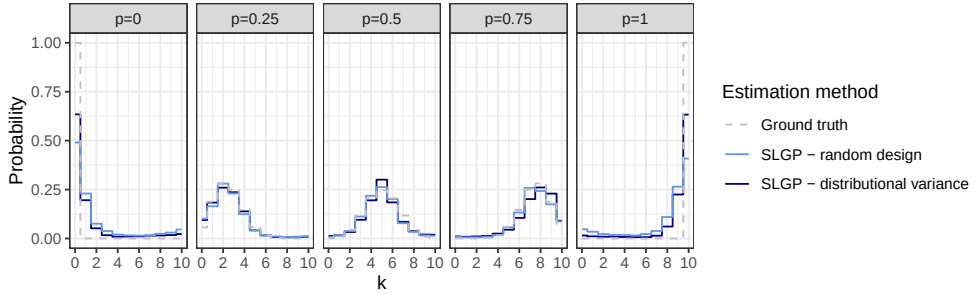


Fig. 5: True binomial distribution, mean SLGP estimate obtained after 200 randomly selected observations, and mean SLGP estimate obtained under the posterior-variance acquisition strategy, displayed for some values of p .

The posterior-variance strategy seems to more accurately reconstruct the distribution near the boundaries of the input domain.

A possible explanation is that for p close to 0 or 1, the binomial distribution concentrates most of its mass near the extreme outcome values, and the pmf varies more rapidly with respect to p . Therefore, small uncertainties and small changes in p may induce substantial changes in the probabilities assigned to the corresponding outcome classes. The SLGP model is able to capture, at least partially, this increased sensitivity and exploit it through the sequential design.

A design that explicitly targets epistemic uncertainty naturally tends to allocate more observations in these regions. This additional sampling effort improves the estimation of both the dominant outcomes and the rapidly decaying tail probabilities,

allowing the SLGP to better recover both the concentration of mass and the sharp decay away from the most likely classes.

In contrast, random sampling does not specifically target these sensitive regions and may therefore provide a less accurate reconstruction of the distributional shape near the boundaries.

Finally, all acquisition strategies exhibit comparable overall computational costs, as illustrated in Figure 6. Both posterior-update costs and the costs associated with selecting the next evaluation point remain at the same order of magnitude across acquisition strategies.

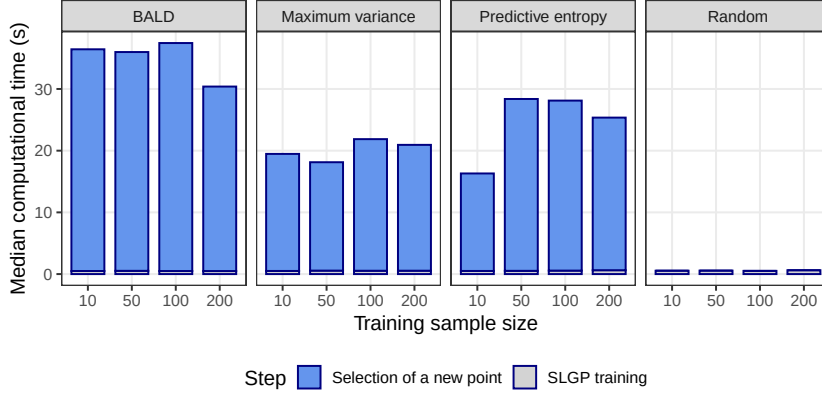


Fig. 6: Median computational cost of the sequential design strategies as a function of the training sample size. Each bar is decomposed into the time required for SLGP training and the time required for acquisition-function optimization. Results are reported for the four acquisition criteria and are aggregated over the 10 Monte Carlo repetitions.

4.1.4 Comparing SLGP to other models

To place the proposed approach in context, we compare SLGP predictions with those obtained from two widely used alternatives for multiclass probability estimation. The first is multiclass Gaussian process classification implemented using the GPflow library [40], relying on sparse variational inference [41]. The second is gradient boosting through the R package `xgboost` [13], using the multiclass objective to directly estimate class probabilities.

Figure 7 reports the evolution of the integrated Kullback-Leibler divergence between the estimated and true distributions and the training time as the number of observations increases. Figure 8 complements this quantitative comparison by displaying representative predictive probability mass functions obtained after 200 observations. The uncertainty ribbons are based on 10 independent repetitions of each experiment.

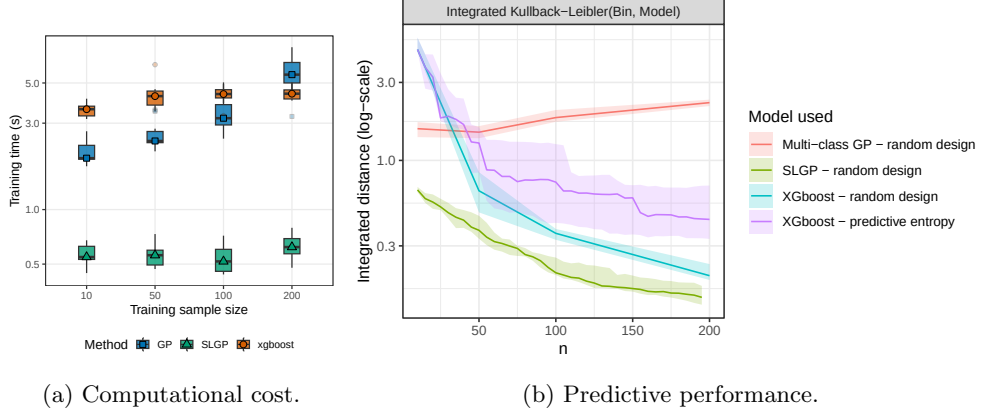


Fig. 7: Comparison of SLGP, XGBoost, and multiclass Gaussian process models. Panel (a) reports the corresponding computational cost. Panel (b) reports the median integrated Kullback–Leibler divergence between the estimated and true distributions as a function of the dataset size under the different acquisition strategies. Shaded ribbons indicate the interquartile range (25%–75% quantiles) across independent repetitions.

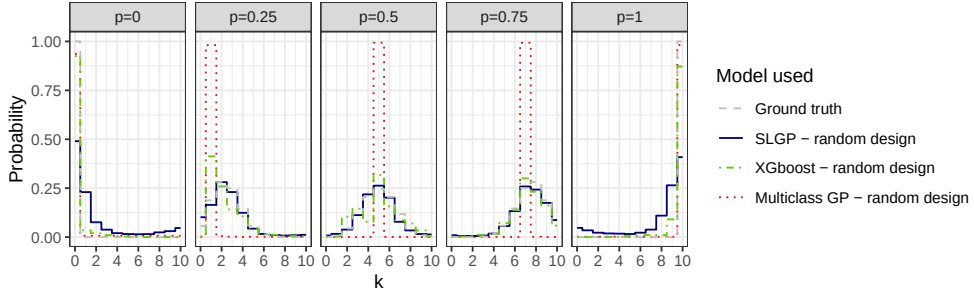


Fig. 8: True binomial distribution, mean SLGP estimate, mean multiclass GP estimate, and XGboost estimate, all obtained after 200 randomly selected observations and displayed for some values of p .

Several observations emerge from Figure 7. First, the proposed SLGP model consistently achieves the lowest integrated Kullback–Leibler divergence. This is perhaps unsurprising given the nature of the underlying data-generating process, which consists of smoothly varying binomial distributions. The SLGP construction is specifically designed to exploit both spatial smoothness and structure on the support through the latent process.

The multiclass GP model remains consistently less accurate. A possible explanation is that standard multiclass GP classification represents categories through distinct

latent processes and does not directly exploit the ordinal structure of the support. Therefore, it is struggling to capture the coexistence of several classes at similar probability levels.

Finally, while XGBoost remains the closest competitor, its performance is noticeably weaker in the present experiment. In particular, the predictive-entropy acquisition strategy performs worse than random sampling. While XGBoost remains a highly flexible predictive model, the present example indicates that flexibility alone is not sufficient in adaptive sampling settings, where accurate uncertainty quantification and the ability to exploit structural assumptions play a central role.

In light of these results and our provided interpretation, we only used SLGP in the upcoming studies, as it was the best performing, even without sequential acquisition.

4.2 Case 2: Binomial distribution with varying n and p

We now extend the analysis to a two-dimensional indexing variable. Although this setting remains moderate in dimension, it introduces additional geometric and structural complexity compared to the one-dimensional case, in particular because the support of the distribution itself varies across the input space.

4.2.1 Data-generating process

We now have a two-dimensional input: $\mathbf{x} = (n_{bin}, p) \in \{5, \dots, 20\} \times (0, 1)$. We denote the data-generating random variables by $T_{\mathbf{x}} = T_{n_{bin}, p}$ in this section. Observations are generated through $T_{n_{bin}, p} \sim \text{Bin}(n_{bin}, p)$, so that the support of $T_{n_{bin}, p}$ is given by $\{0, \dots, n_{bin}\} \subset \{0, \dots, 20\}$. We model it with a SLGP, and the number of categories is $K = 21$.

The conditional probability mass function is

$$\mathbb{P}(T_{n_{bin}, p} = k) = \mathbb{1}_{k \leq n_{bin}} \binom{n_{bin}}{k} p^k (1-p)^{n_{bin}-k}, \quad k = 0, \dots, 20. \quad (23)$$

This example defines a parametric family of unimodal discrete distributions indexed by n_{bin}, p , with varying skewness and support across the domain. Figure 9 visualizes the resulting probability mass functions as a function of the parameters n_{bin}, p and outcome k .

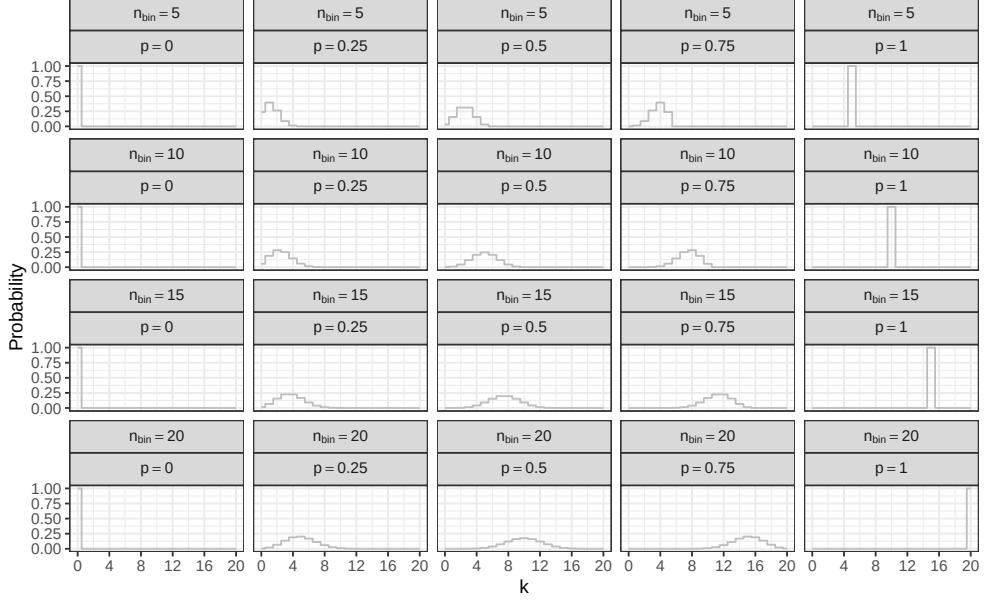


Fig. 9: Ground-truth binomial probability mass function for some n_{bin} and p values.

To model this family, as in the previous section, we employ a finite-rank SLGP. The basis functions are constructed using a Random Fourier Feature (RFF) approximation of a Matérn-5/2 kernel. We sample 100 frequencies, yielding 200 sine and cosine basis functions.

In contrast to the previous example, the effective support and shape of the distribution vary more substantially across the input space. This makes the inference and sequential design more involved.

4.2.2 Performance metrics

Because the target distribution has a support that is often strictly included in $\{0, \dots, 21\}$, we replace the KL divergence by a metric that is more robust to support shifts. We use the Wasserstein-1 distance (earth mover’s distance), defined for two CDFs of discrete probability mass functions F and G with common support $\{0, \dots, K\}$ as: $W_1(F, G) = \sum_{k=0}^K |F(k) - G(k)|$. Unlike KL divergence, W_1 remains stable when the support of the distributions changes, making it more appropriate for our case.

4.2.3 Sequential design of experiments

We repeat the sequential acquisition experiment under the same protocol as in the one-dimensional case. We do not use a mixed-integer optimiser for the acquisition criterion optimisation, but rather perform a grid search with an optimisation over p for each value of $n_{bin} \in \{5, \dots, 20\}$ and keeping the best couple of n_{bin_new}, p_{new} . We

then add 5 new trial drawn from $\text{Bin}(n_{\text{bin new}}, p_{\text{new}})$ to the dataset. Once again, the choice of five observations is primarily motivated by simplicity.

Figure 10 reports the evolution of the integrated Wasserstein-1 distance as a function of dataset size.

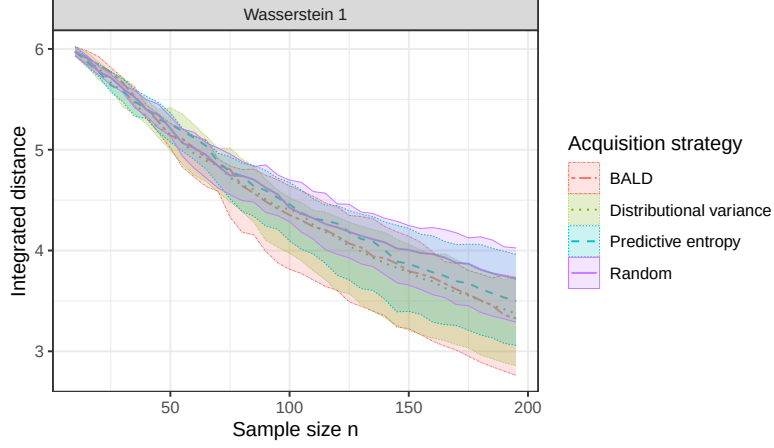


Fig. 10: Median integrated Wasserstein-1 distance between the SLGP prediction and the ground-truth binomial model as a function of dataset size under different acquisition strategies. Shaded ribbons indicate the interquartile range (25%-75% quantiles) across independent repetitions.

In contrast with the one-dimensional experiment, no acquisition strategy clearly dominates in this two-dimensional setting.

Figure 11 provides a panel comparison of the true binomial distribution, the SLGP estimate obtained after 200 randomly selected observations, and the SLGP estimate obtained under the BALD acquisition strategy. In both cases, the model struggles to accurately capture the abrupt structural changes induced by variations in n , particularly the expansion and contraction of the effective support.

This behaviour suggests that, in the absence of explicit structural information, purely smooth latent representations may have difficulty adapting to discontinuous support transitions.

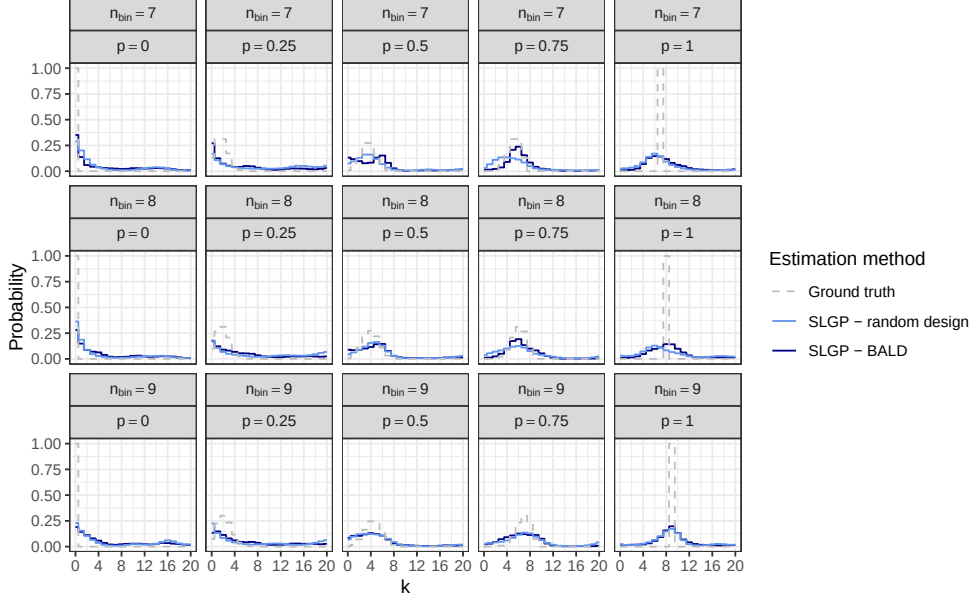


Fig. 11: True binomial distribution, mean SLGP estimate obtained after 200 randomly selected observations, and mean SLGP estimate obtained under the BALD acquisition strategy, displayed for some values of n_{bin} , p .

4.2.4 Introducing a structured trend to account for support variation

To better accommodate the systematic support changes induced by varying n_{bin} , we incorporate structural prior knowledge into the latent energy surface. Specifically, we introduce a deterministic trend component in the latent Gaussian process defined as

$$\mu(n_{bin}, p, k) = -10 \cdot \mathbb{1}_{\{k > n_{bin}\}},$$

which penalises outcome categories that lie outside the effective support $\{0, \dots, n_{bin}\}$. Indeed, looking back at Equation 6, we have a SLGP model where:

$$Y_{n_{bin}, p}(k) = \frac{\exp \{-10 \cdot \mathbb{1}_{\{k > n_{bin}\}} + \varepsilon^\top \mathbf{F}(n_{bin}, p, k)\}}{\sum_{j=0}^K \exp \{-10 \cdot \mathbb{1}_{\{j > n_{bin}\}} + \varepsilon^\top \mathbf{F}(n_{bin}, p, j)\}},$$

and as such:

$$Y_{n_{bin}, p}(k) \propto \begin{cases} \exp \{\varepsilon^\top \mathbf{F}(n_{bin}, p, k)\}, & \text{if } k \leq n_{bin}, \\ \exp \{-10 + \varepsilon^\top \mathbf{F}(n_{bin}, p, k)\} \approx 4.5 \times 10^{-5} \exp \{\varepsilon^\top \mathbf{F}(n_{bin}, p, k)\}, & \text{if } k > n_{bin}. \end{cases}$$

This structured component encourages the model to respect the natural support constraint while retaining flexibility through the residual GP term.

We repeat the sequential acquisition experiment under the same protocol as before. Figure 12 illustrates the evolution of the integrated Wasserstein-1 distance.

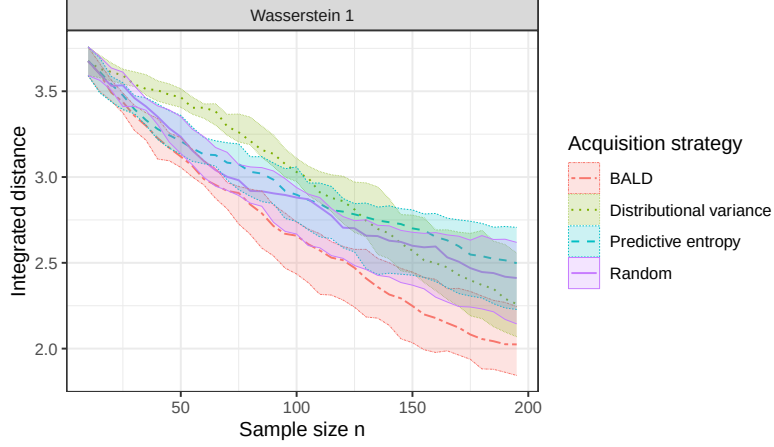


Fig. 12: Median integrated Wasserstein-1 distance between the SLGP prediction and the ground-truth binomial model as a function of dataset size under different acquisition strategies. Shaded ribbons indicate the interquartile range (25%-75% quantiles) across independent repetitions.

The inclusion of the structural trend substantially alters the relative performance of the acquisition strategies. BALD now clearly outperforms the alternatives. The posterior-variance criterion initially exhibits slower improvement but progressively approaches the performance of other methods as the dataset grows.

Although residual discrepancies remain, the introduction of the trend significantly improves the reconstruction of support transitions across n . This confirms that explicitly encoding structural constraints reduces the burden on the purely smooth latent component and enables more efficient information acquisition.

Figure 13 provides a panel comparison of the true binomial distribution, the SLGP estimate obtained after 200 randomly selected observations, and the SLGP estimate obtained under the BALD acquisition strategy.

While the model still encounters difficulty near extreme configurations (e.g., p close to 0 or 1, and near the boundary $k \approx n_{bin}$), the structural trend markedly reduces systematic support misalignment. Moreover, BALD-driven acquisition visibly improves reconstruction quality compared to random sampling, both quantitatively and qualitatively.

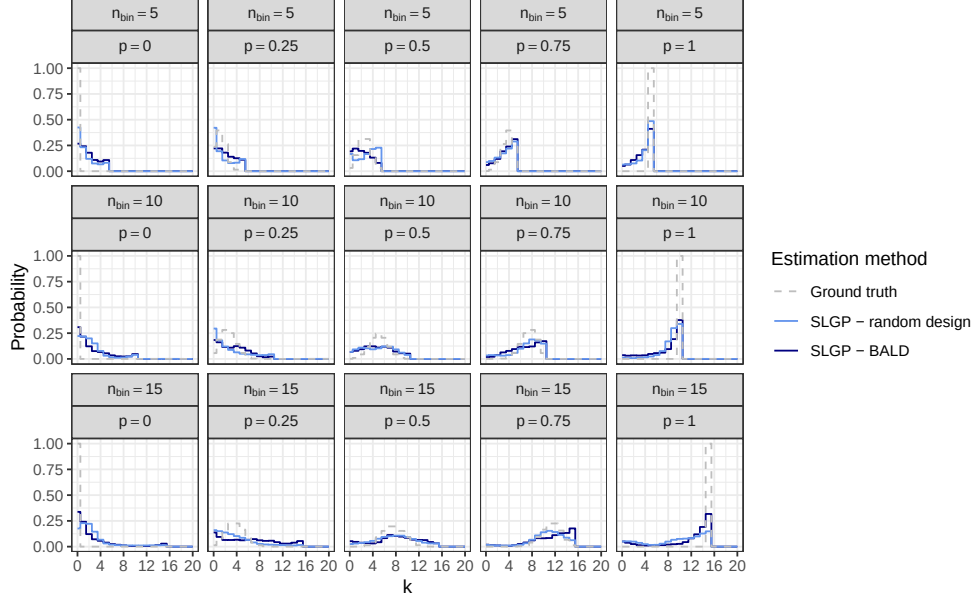


Fig. 13: True binomial distribution, mean SLGP estimate obtained after 200 randomly selected observations, and mean SLGP estimate obtained under the BALD acquisition strategy, displayed for some values of n_{bin}, p .

Finally, the computational times are larger in this second case, reflecting the increased complexity of the underlying distributional model. However, the overall behaviour remains qualitatively similar to that observed in Case 1. As illustrated in Figure 14, both the posterior-update cost and the acquisition-function optimization cost remain comparable across the different acquisition strategies. No strategy appears to induce a substantially larger computational burden than the others.

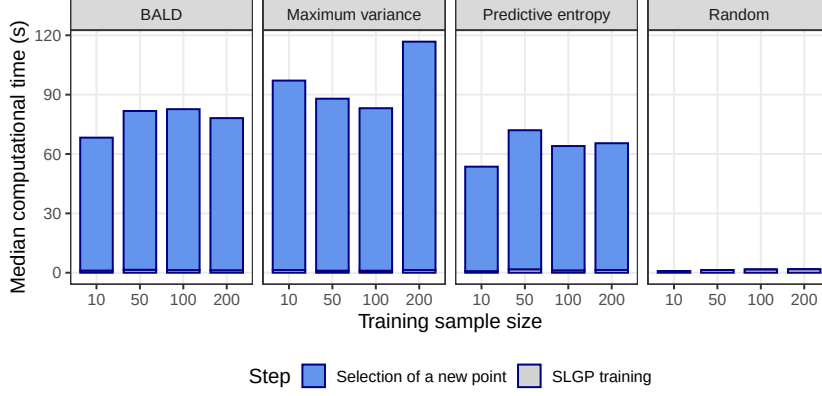


Fig. 14: Median computational cost of the sequential design strategies as a function of the training sample size. Each bar is decomposed into the time required for SLGP training and the time required for acquisition-function optimization. Results are reported for the four acquisition criteria and are aggregated over the 10 Monte Carlo repetitions.

4.3 Real-data application: *Bombus terrestris* occurrences in France

To illustrate the practical applicability of the proposed framework on real data, we consider occurrence records from the Global Biodiversity Information Facility (GBIF) [42, 43]. The complete extraction and preprocessing pipeline (including GBIF queries and spatial aggregation) is available in the accompanying GitHub repository.

4.3.1 Data extraction and preprocessing

We collect occurrence records of *Bombus terrestris* (bumblebee) in metropolitan France over the period 2019 - 2023, retaining only observations with valid geographic coordinates and temporal information. The indexing variable is spatial, $\mathbf{x} = (\text{lon}, \text{lat}) \in D \subset \mathbb{R}^2$.

To ensure computational tractability while preserving spatial resolution, we discretize the French territory using a regular hexagonal tessellation. Each hexagon defines a spatial cell h . Hexagonal cells are preferred over square grids due to their isotropic neighbourhood structure and reduced directional bias.

$$t_{h,y} = \min(50, \# \{ \text{occurrences of } \textit{Bombus terrestris} \text{ falling in } h \text{ during year } y \})$$

This construction yields a nonnegative integer-valued response defined on a common spatial support across years, suitable for discrete-output SLGP modelling. We also apply an upper truncation to the response to mitigate the influence of extreme counts, which are typically associated with densely populated or intensively sampled areas.

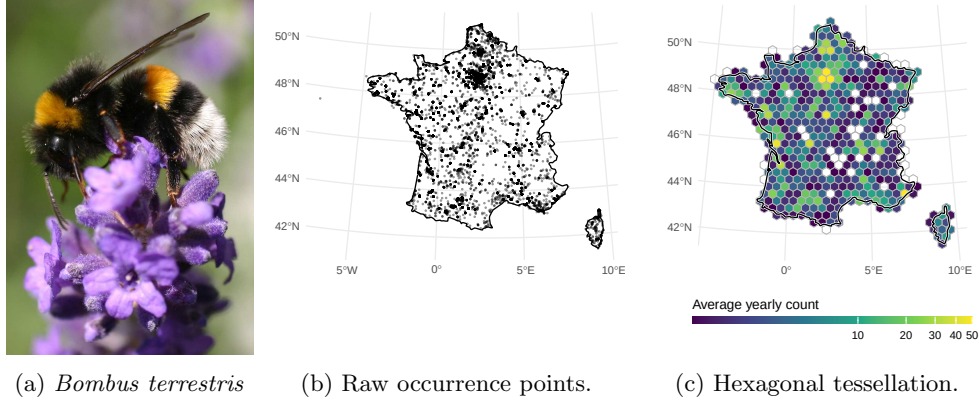


Fig. 15: Spatial distribution of *Bombus terrestris* occurrences and hexagonal discretization of the spatial domain, total counts over the period 2019 - 2023.

The resulting dataset can be interpreted as a spatio-temporal collection of integer-valued observations with fixed spatial locations (e.g., hexagon centroids) and repeated measurements over time.

4.3.2 Training and evaluation protocol

The training set consists of the count observations for year 2020 to 2021 at the centroid of each hexagonal cell. Each year is treated as an independent spatial replication, and as before, the SLGP finite rank representation leverages a RFF approximation of a Matérn-5/2 kernel. We sample 200 frequencies, yielding 400 sine and cosine basis functions.

To assess predictive performance in a distributional sense, we compare SLGP-based predictions with empirical summaries computed from an extended reference window covering 2019 - 2023. This longer period provides a richer empirical approximation of the underlying spatial count distribution at each cell.

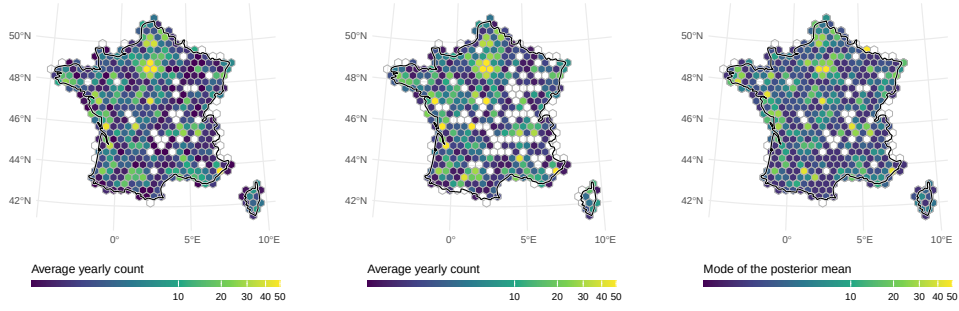
More precisely, for each hexagon h , we compute the empirical average yearly count $\bar{t}_h^{\text{ref}} = \frac{1}{5} \sum_{y=2019}^{2023} t_{h,y}$, which serves as a long-run reference summary.

We then contrast this quantity with the mode of the posterior predictive distribution inferred by the SLGP model fitted on the restricted training period 2020-2021. This comparison evaluates the ability of the model, trained on limited data, to recover the dominant spatial patterns.

4.3.3 Visualization and qualitative assessment

We provide a spatial comparison between empirical long-run averages and SLGP-based predictive summaries.

While this evaluation remains qualitative, these visuals provide insight into the capacity of the SLGP framework to recover large-scale spatial structure from limited information.



(a) Average yearly count: full dataset (2019-2023). (b) Average yearly count: training dataset (2020-2021). (c) Mode of the posterior predictive distribution.

Fig. 16: Spatial comparison between empirical summaries and the SLGP-based predictive summary.

5 Conclusion

We have introduced a Spatial Logistic Gaussian Process (SLGP) framework tailored to the modelling of spatially indexed discrete responses. By extending the continuous SLGP construction to finite-support outcomes, the proposed approach enables flexible, nonparametric modelling of spatially varying count distributions while preserving full distributional information and coherent posterior uncertainty quantification. We demonstrated the applicability of SLGP modelling to sequential data acquisition on a reduced benchmark. Through controlled synthetic experiments, we demonstrated that epistemic-uncertainty-driven acquisition criteria—such as BALD and posterior variance of class probabilities—substantially improve distributional reconstruction efficiency compared to random sampling and predictive entropy. These results provide a proof of concept that SLGPs can be embedded in an adaptive design loop where full predictive distributions guide data collection.

The real-data application to *Bombus terrestris* occurrences over metropolitan France further illustrates the practical relevance of the framework. Despite being trained on restricted data, the SLGP model recovers spatial patterns. The ability to model the entire predictive pmf, rather than only conditional means, will prove particularly valuable in heterogeneous ecological settings.

Several limitations and extensions naturally arise from the present work.

First, the trend function $\mu(x, t)$ is assumed to be specified by the practitioner and is not estimated from the data. Joint inference on trend parameters and latent Gaussian components could provide additional modelling flexibility while preserving the probabilistic structure of the model.

Second, the implementation currently relies on MCMC or Laplace approximation, while the connections between discrete-output SLGPs, multiclass Gaussian process classification, and Bayesian multinomial models suggest opportunities to leverage more advanced variational or augmentation-based inference schemes, particularly when the support size becomes large.

Third, the active-learning strategies considered in this work are based on posterior uncertainty measures, and while the present work establishes a proof of concept for entropy, variance, and mutual-information-based acquisition criteria, richer approaches such as Stepwise Uncertainty Reduction (SUR) [44] could be adapted to directly target reduction of uncertainty on distributional functionals (e.g., modes, quantiles, spatial integrals). Extending such criteria to discrete and continuous SLGPs would provide a theoretically grounded design layer tailored to distributional objectives. Likewise, the current sequential design setting assumes a fixed number of observations acquired at each iteration. Jointly optimizing both the sampling location and the sampling effort would constitute a natural extension of the proposed framework.

Finally, in our current work, performance assessment is either on analytical data through distance to the known data generating process, or mostly visual. It would be interesting to be able to methodically assess distributional predictive performance in situations where the true underlying distribution is unknown, motivating a more systematic investigation of proper scoring rules and probabilistic forecast evaluation techniques [45].

Overall, this work demonstrates that SLGPs constitute a flexible, computationally feasible, and acquisition-aware framework for modelling spatially varying discrete distributions, and open promising perspectives for adaptive spatial data collection and uncertainty-aware ecological monitoring.

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